

Definition 2.8: Cumulative distribution function (CDF) of a finite list

The CDF of $z \in \mathbb{R}^n$ is the function $\hat{F}_z : \mathbb{R} \rightarrow [0, 1]$ defined as

$$\hat{F}_z(v) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{z_i \leq v\}.$$

Definition 2.9: Quantile of a finite list

For any $\tau \in [0, 1]$, the τ -quantile of $z \in \mathbb{R}^n$ is defined as

$$\text{Quantile}(z; \tau) = \inf \left\{ v : \hat{F}_z(v) \geq \tau \right\}.$$

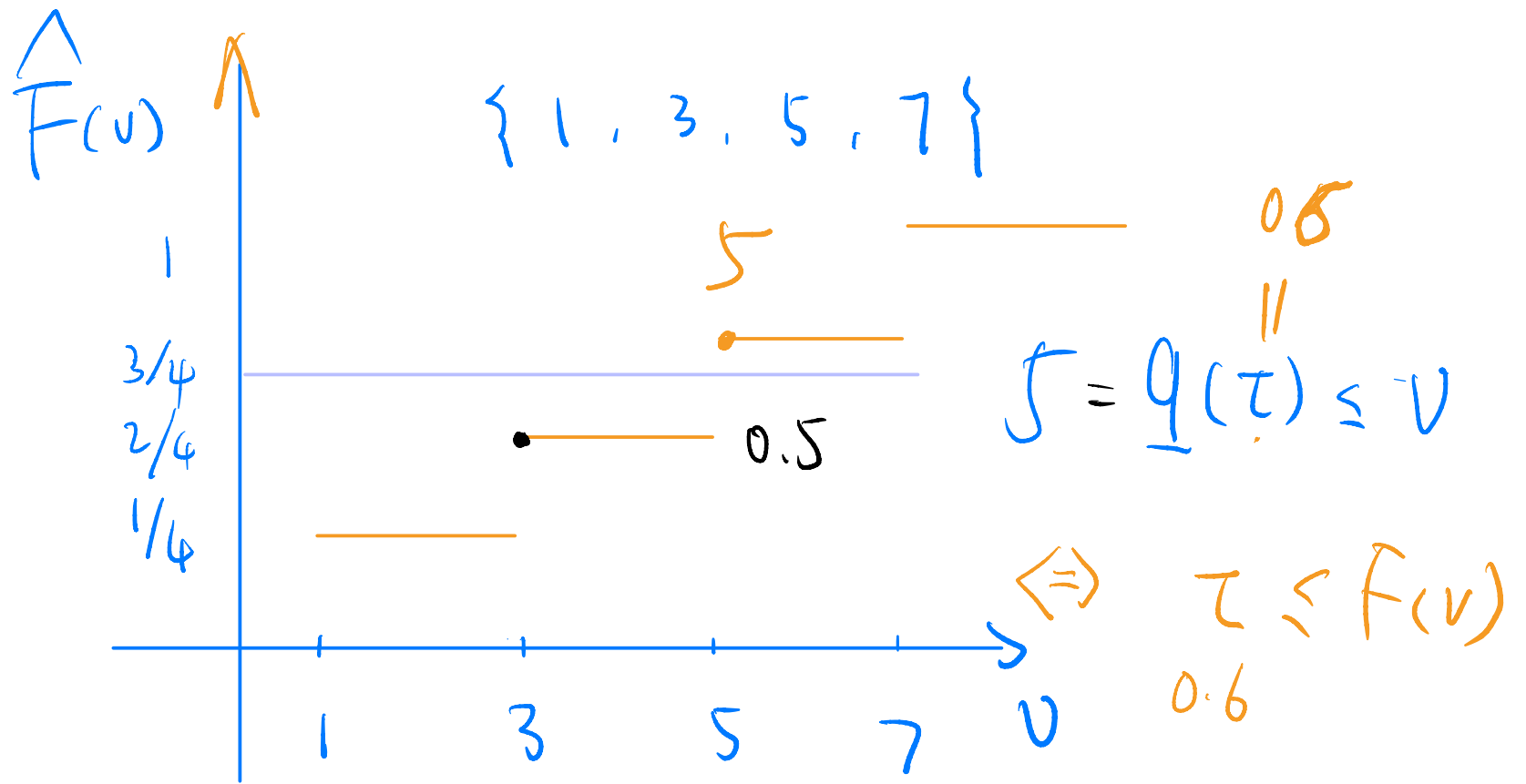
Definition 2.7: Order statistics of a finite list

Let $k \in [n]$. Then the k th order statistic of $z \in \mathbb{R}^n$, written as $z_{(k)}$, is defined as

$$z_{(k)} = \inf \left\{ v : \sum_{i=1}^n \mathbb{1}\{z_i \leq v\} \geq k \right\}.$$

$\{1, 3, 5, 7\}$

$\tau(2) = 3$



$$\hat{F}(v) = \frac{\sum_{i=1}^n \mathbb{1}(z_i \leq v)}{n}$$

$$\text{Quantile}(z, 0.5) = \inf \{ z : \hat{F}(z) \geq 0.5 \} = 3$$

$0.6 = 5$

Fact 2.10: Conversion between order statistics and quantiles.

For any $z \in \mathbb{R}^n$, for all $\tau \in (0, 1]$,

$$z_{(\lceil \tau n \rceil)} = \text{Quantile}(z; \tau).$$

Proof. For $v \in \mathbb{R}$, write

$$N_z(v) = \sum_{i=1}^n \mathbf{1}\{z_i \leq v\}, \quad \widehat{F}_z(v) = \frac{N_z(v)}{n}.$$

Because $N_z(v)$ is integer-valued,

$$\widehat{F}_z(v) \geq \tau \iff N_z(v) \geq n\tau \iff N_z(v) \geq \lceil n\tau \rceil.$$

Therefore, using the book's definitions of quantiles and order statistics,

$$\begin{aligned} \text{Quantile}(z; \tau) &= \inf\{v : \widehat{F}_z(v) \geq \tau\} \\ &= \inf\{v : N_z(v) \geq \lceil n\tau \rceil\} \\ &= z_{(\lceil n\tau \rceil)}. \end{aligned}$$

Here $0 < \tau \leq 1$ ensures that $\lceil n\tau \rceil \in \{1, \dots, n\}$. The argument also allows ties among the z_i . $\square$

$$R_i = |y_i - f(x_i)| \quad i = 1, \dots, n$$

$$Q_n = \underline{\tau_{(1-\alpha)(n+1)}} \text{ of } R_1, \dots, R_{n+1}$$

$$= R_{(\tau_{(1-\alpha)(n+1)})}$$

$$= \text{quantile}(R, 1-\alpha)$$

Fact 2.11: Conversion between order statistics and CDFs.

For any $z \in \mathbb{R}^n$, for all $k \in [n]$,

$$\hat{F}_z(z_{(k)}) \geq \frac{k}{n},$$

with equality in the case that all elements of z are distinct.

Write the observations in increasing order:

$$z_{(1)} \leq z_{(2)} \leq \dots \leq z_{(n)}.$$

By definition, $\hat{F}_z(z_{(k)})$ is the fraction of observations at or below $z_{(k)}$:

$$\hat{F}_z(z_{(k)}) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{z_i \leq z_{(k)}\}.$$

The first k ordered observations are all at or below $z_{(k)}$. Thus at least k observations are counted, giving

$$\hat{F}_z(z_{(k)}) \geq \frac{k}{n}.$$

If all observations are distinct, every observation after the k th is strictly greater than $z_{(k)}$. Then exactly k observations are counted, so equality holds.

With ties, additional observations can be counted. For example, if $z = (1, 2, 2, 3)$ and $k = 2$, then

$$z_{(2)} = 2.$$

$$\hat{F}(2) = 3 \geq 2$$

Fact 2.12: Conversion between quantiles and CDFs.

The following equivalences hold for any $z \in \mathbb{R}^n$:

- (i) $\hat{F}_z(v) = \sup\{\tau : \text{Quantile}(z; \tau) \leq v\}$ for all $v \in \mathbb{R}$;
- (ii) $\text{Quantile}(z; \hat{F}_z(v)) \leq v$ for all $v \in \mathbb{R}$;
- (iii) $\hat{F}_z(\text{Quantile}(z; \tau)) \geq \tau$ for all $\tau \in [0, 1]$;

and furthermore, as a special case of (iii),

- (iv) If all elements of z are distinct, $\hat{F}_z(\text{Quantile}(z; \tau)) = \frac{\lceil \tau n \rceil}{n}$ for all $\tau \in [0, 1]$.

Write $F(v) = \hat{F}_z(v)$ and $Q(\tau) = \text{Quantile}(z; \tau)$. Recall the book's conventions

$$Q(0) = -\infty, \quad F(-\infty) = 0.$$

The key relationship is

$$Q(\tau) \leq v \iff \tau \leq F(v). \quad (*)$$

It says that the τ -quantile lies at or below v exactly when at least a fraction τ of the observations lie at or below v .

To prove this for $\tau > 0$, let $m = \#\{i : z_i \leq v\}$. By Fact 2.10,

$$\begin{aligned} Q(\tau) \leq v &\iff z_{(\lceil n\tau \rceil)} \leq v \\ &\iff \lceil n\tau \rceil \leq m \\ &\iff n\tau \leq m \\ &\iff \tau \leq F(v). \end{aligned}$$

The second equivalence holds because the k th smallest observation is at or below v exactly when at least k observations are at or below v . For $\tau = 0$, both sides of $(*)$ are true.

$$Q(\tau) \leq v \iff \tau \leq F(v)$$

(i) By (*), the quantile levels whose quantiles lie at or below v are precisely

$$\{\tau \in [0, 1] : Q(\tau) \leq v\} = [0, F(v)].$$

Taking the supremum gives

$$F(v) = \sup\{\tau \in [0, 1] : Q(\tau) \leq v\}.$$

(ii)

We already proved that, for any $\tau \in [0, 1]$,

$$Q(\tau) \leq v \iff \tau \leq F(v).$$

Now fix v . Since $F(v)$ is a number in $[0, 1]$, we can choose it as our quantile level: $\tau = F(v)$. Substituting this into **both sides** gives

$$Q(F(v)) \leq v \iff F(v) \leq F(v).$$

The right-hand statement is always true, so the left-hand statement must also be true:

$$Q(F(v)) \leq v.$$

Intuitively, $F(v)$ tells us the fraction of observations at or below v . Then $Q(F(v))$ finds the **smallest threshold needed to reach that fraction**. Since the threshold v already reaches it, the smallest such threshold cannot exceed v .

For example, take

$$z = (1, 3, 7, 9), \quad v = 5.$$

(iii) For $\tau > 0$, let $k = \lceil n\tau \rceil$. Facts 2.10 and 2.11 give

$$F(Q(\tau)) = F(z_{(k)}) \geq \frac{k}{n} = \frac{\lceil n\tau \rceil}{n} \geq \tau.$$

At $\tau = 0$, $F(Q(0)) = F(-\infty) = 0$. Thus

$$F(Q(\tau)) \geq \tau \quad \text{for all } \tau \in [0, 1].$$

(iv) If all observations are distinct, exactly k observations lie at or below $z_{(k)}$. Therefore, for $\tau > 0$,

$$F(Q(\tau)) = F(z_{(\lceil n\tau \rceil)}) = \frac{\lceil n\tau \rceil}{n}.$$

At $\tau = 0$, both sides equal zero, so the formula holds on all of $[0, 1]$.

Fact 2.13: Order statistics, quantiles, and CDFs under transformations

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a monotone nondecreasing function. Furthermore, for $z \in \mathbb{R}^n$, let $f(z) \in \mathbb{R}^n$ denote the elementwise application of f to z . Then,

- (i) For all $k \in [n]$, $f(z)_{(k)} = f(z_{(k)})$.
- (ii) For all $\tau \in (0, 1]$, $\text{Quantile}(f(z); \tau) = f(\text{Quantile}(z; \tau))$.
- (iii) If additionally f is a strictly increasing function, then for all $v \in \mathbb{R}$, $\hat{F}_{f(z)}(f(v)) = \hat{F}_z(v)$.

Fact 2.14: Properties of order statistics, quantiles, and CDFs under permutations

For any vector $z \in \mathbb{R}^n$ and any permutation σ on $[n]$, let $z_\sigma = (z_{\sigma(1)}, \dots, z_{\sigma(n)})$, i.e., the entries of z are permuted according to σ . We have that

- (i) For all $k \in [n]$, $(z_\sigma)_{(k)} = z_{(k)}$.
- (ii) For all $\tau \in [0, 1]$, $\text{Quantile}(z_\sigma; \tau) = \text{Quantile}(z; \tau)$.
- (iii) For all $v \in \mathbb{R}$, $\hat{F}_{z_\sigma}(v) = \hat{F}_z(v)$.

Fact 2.15: Order statistics, quantiles, and CDFs under exchangeability

Assume $Z \in \mathbb{R}^n$ is exchangeable, and fix any $i \in [n]$. Then we have that

- (i) For any $k \in [n]$, $\mathbb{P}(Z_i \leq Z_{(k)}) \geq k/n$ and $\mathbb{P}(Z_i < Z_{(k)}) \leq (k-1)/n$.
- (ii) For all $\tau \in [0, 1]$, $\mathbb{P}(Z_i \leq \text{Quantile}(Z; \tau)) \geq \tau$ and, if $\tau > 0$, $\mathbb{P}(Z_i < \text{Quantile}(Z; \tau)) < \tau$.
- (iii) For all $\tau \in [0, 1]$, $\mathbb{P}(\hat{F}_Z(Z_i) \leq \tau) \leq \tau$ and $\mathbb{P}(\hat{F}_Z(Z_i) \geq \tau) \geq 1 - \tau$.

Furthermore, if all elements of Z are distinct almost surely,

- (iv) For any $k \in [n]$, $\mathbb{P}(Z_i \leq Z_{(k)}) = k/n$.
- (v) For all $\tau \in [0, 1]$, $\mathbb{P}(Z_i \leq \text{Quantile}(Z; \tau)) = \frac{\lfloor n\tau \rfloor}{n}$.
- (vi) For all $\tau \in [0, 1]$, $\mathbb{P}(\hat{F}_Z(Z_i) \leq \tau) = \frac{\lfloor n\tau \rfloor}{n}$.

$$R_i = |y_i - f(x_i)| \quad i=1 \dots n$$

$$R_{n+1} = \underbrace{|y_{n+1} - f(x_{n+1})|}_{?}$$

$$R_{n+1} \leq \underbrace{\tau_{(1-\alpha)(n+1)}}_k \text{-th smallest of } R_1 \dots R_{n+1}$$

$\Leftrightarrow$

$$R_{n+1} \leq k \text{-th smallest of } R_1 \dots R_n$$

$$\mathbb{P}(R_{n+1} \leq k \text{-th smallest of } R_1 \dots R_n)$$

$$\underline{P(R_{n+1} \leq R_{(k)}) \geq 1 - \alpha}$$

Fact 2.15: Order statistics, quantiles, and CDFs under exchangeability

Assume $Z \in \mathbb{R}^n$ is exchangeable, and fix any $i \in [n]$. Then we have that

- (i) For any $k \in [n]$, $\mathbb{P}(Z_i \leq Z_{(k)}) \geq k/n$ and $\mathbb{P}(Z_i < Z_{(k)}) \leq (k-1)/n$.
- (ii) For all $\tau \in [0, 1]$, $\mathbb{P}(Z_i \leq \text{Quantile}(Z; \tau)) \geq \tau$ and, if $\tau > 0$, $\mathbb{P}(Z_i < \text{Quantile}(Z; \tau)) < \tau$.
- (iii) For all $\tau \in [0, 1]$, $\mathbb{P}(\widehat{F}_Z(Z_i) \leq \tau) \leq \tau$ and $\mathbb{P}(\widehat{F}_Z(Z_i) \geq \tau) \geq 1 - \tau$.

Furthermore, if all elements of Z are distinct almost surely,

- (iv) For any $k \in [n]$, $\mathbb{P}(Z_i \leq Z_{(k)}) = k/n$.
- (v) For all $\tau \in [0, 1]$, $\mathbb{P}(Z_i \leq \text{Quantile}(Z; \tau)) = \frac{\lceil n\tau \rceil}{n}$.
- (vi) For all $\tau \in [0, 1]$, $\mathbb{P}(\widehat{F}_Z(Z_i) \leq \tau) = \frac{\lfloor n\tau \rfloor}{n}$.